# Zhangqi (Gavin) Duan

zduan@cs.umass.edu | Website | LinkedIn | Google Scholar

Amherst, MA

Summary

PhD candidate in Computer Science at UMass Amherst specializing in LLM training and alignment, with a focus on education technology and human-centered AI systems. Experienced in LLM fine-tuning (SFT), RL–based optimization (GRPO, DPO), and multitask training for interpretable model behavior. Published in leading venues (ACL, EMNLP, AIED) with a strong record of translating research prototypes into robust ML systems for real-world NLP and reasoning tasks.

## Education

**University of Massachusetts Amherst**

- PhD in Computer Science (Advisor: Prof. Andrew Lan), GPA: 3.83/4.00 — Sep 2024 – Present
- MS in Computer Science, GPA: 3.82/4.00 — Sep 2022 – May 2024
- BS in Computer Science & BS in Mathematics, GPA: 3.85/4.00 — Sep 2018 – May 2022

## Work Experience

**UMass Amherst Center for Data Science** — Amherst, MA

Machine Learning Engineer Intern — May 2023 – Aug 2023

- Analyzed 6.5M tweets using scalable data processing methods to detect key patterns and outliers
- Applied Python Numpy, Pandas modules to perform advanced data cleaning and feature engineering resulted in 23% decrease in model runtime
- Built a Topic Model to categorize diverse topics, uncovering unknown events within Non-English textual data
- Implemented a User Model pipeline to target primary contributors to ongoing conversations with unsupervised learning algorithms (Spherical K-means clustering, hierarchical clustering)

**Siemens Ltd** — Beijing, China

Software Engineer Intern — May 2019 - Aug 2019

- Conducted database creation for 10,000 clients using SQL Subqueries/Merge Statement
- Resolved technical issues reported by clients or detected through internal monitoring
- Performed code reviews and provided feedback to team members

## Publications

**Who Am I? History-Aware Profiles for Student Simulation in Tutoring Dialogues**
**Zhangqi Duan**, Shuyan Huang, Alexander Scarlatos, Jaewook Lee, Simon Woodhead, Andrew Lan
*EMNLP, 2026*

**Knowledge-Aligned Student Error Simulator for Open-Ended Coding Tasks**
**Zhangqi Duan**, Nigel Fernandez, Andrew Lan
*ACL, 2026*

**Automated Knowledge Component Generation for Interpretable Knowledge Tracing in Coding Problems**
**Zhangqi Duan**, Nigel Fernandez, Arun Narayanan, Mohammad Hassany, Rafaella Alencar, Peter Brusilovsky, Bita Akram, Andrew Lan
*ACL Findings, 2026*

**Using LLMs for Knowledge Component-level Correctness Labeling in Open-ended Coding Problems**

**Zhangqi Duan**, Arnav Kankaria, Dhruv Kartik, Andrew Lan
*AIED, Short Paper, 2026*

**Test Case-Informed Knowledge Tracing for Open-ended Coding Tasks**
**Zhangqi Duan**, Nigel Fernandez, Alexander Hicks, Andrew Lan
*LAK, 2025*

**Multilingualism, Transnationality, and K-pop in the Online #StopAsianHate Movement**
Tessa Masis, **Zhangqi Duan**, Weiai Wayne Xu, Ethan Zuckerman, Jane Yeahin Pyo, Brendan O'Connor
*WebSci, 2025*

**README: Bridging Medical Jargon and Lay Understanding for Patient Education through Data-Centric NLP**
Zonghai Yao, Nandyala Siddharth Kantu, Guanghao Wei, Hieu Tran, **Zhangqi Duan**, Sunjae Kwon, Zhichao Yang, README annotation team, Hong Yu
*EMNLP Findings, 2024*

**Agent Performing Autonomous Stock Trading under Good and Bad Situations**
Yunfei Luo, **Zhangqi Duan**
*AI4ABM (ICLR Workshop), 2023*

## Research Experience

**Persona-Grounded User Simulation for Tutoring Dialogue** Feb 2026 – May 2026

- Formulated a history-aware student simulation task that models tutoring dialogue from both student knowledge and behavioral perspectives
- Developed a two-component **RL-based** framework for student simulation in tutoring dialogue, improving faithfulness by **4.2%** in Act Matching and **3.8%** in Error Matching
- Devised a **counterfactual DPO preference-pair construction mechanism** for the student simulator, enabling better persona alignment
- Optimized the persona generator through a custom **GRPO** reward function, refining student persona to improve downstream simulation performance

**LLM Post-Training for Code Generation** Oct 2025 – Jan 2026

- Developed **RL-based** methodology **(GRPO)** for student code simulation, improving controllability and diversity of generated codes
- Designed a custom **RL reward** function combining code similarity metrics and error level IoU to better align generation with student error distributions
- Evaluated SFT-only vs. SFT+RL training at scale, yielding a **13.1%** improvement in error coverage
- Built an end-to-end LLM training and evaluation pipeline in PyTorch, supporting reproducible experimentation across reward variants and datasets

**Multitask LLM Training for Interpretable Prediction and Generation** Aug 2024 – Jan 2025

- Created the first LLM-based pipeline for knowledge component generation, validated against expert-level quality
- Designed a **soft-token conversion mechanism** to support end-to-end gradient flow through multiple models
- Developed KC-Gen KT, a **multitask learning** framework jointly optimizing code generation and correctness prediction, integrating interpretable student representations
- Achieved a **5%** AUC improvement on prediction tasks and a **4%** CodeBLEU gain on code generation tasks over state-of-the-art models, demonstrating the effectiveness of KC-Gen KT across diverse tasks

**Retrieval-Augmented Lay Definition Generation in BioNLP** Sep 2021 – Jan 2022

- Improved patient comprehension of clinical notes using natural language generation
- Applied pretrained sequence-to-sequence models (BART) for controlled definition generation
- Built data pipelines to handle linguistic ambiguity and domain shift in medical text
- Deployed a full-stack NLP application using React and Django to deliver real-time explanations to end users

Teaching Experience

**Teaching assistant, College of Info. and Computer Science** Fall 2022, Spring 2023

CS 240: Reasoning Under Uncertainty

- Assigned different tasks such as checking assignment and quiz correctness to other TAs and UCAs as head TA
- Assisted in preparation of coursework and taught discussion sections, improving the learning experience for 200+ undergraduate students

Service

**Mentoring** Sep 2025 – Present

Two undergraduate students, resulting in a publication at AIED 2026

**Program Chair Member** at AIED 2026

Honors & Awards

Bay State Fellowship (Full Scholarship Sponsored by UMass CICS) 2022
Graduated Magna Cum Laude In MS from University of Massachusetts Amherst
Graduated Magna Cum Laude In BS from University of Massachusetts Amherst
Dean's List in all semesters

Skills

**Machine Learning**: deep neural network architecture design, reinforcement learning algorithm implementation and reward/loss function design, preference alignment
**Large Language Models**: prompt engineering and RAG, LLM-driven annotation and evaluation, fine-tuning and LoRA using SFT, DPO, GRPO
**Software Engineering**: full-stack web development
**Programming Languages**: Python, Java, JavaScript, HTML, CSS, SQL
**Frameworks**: PyTorch, HuggingFace, TRL, vLLM, React, Node.js